ChIP-seq Data Analysis Assignment

Experiment: Histone modification ChIP-seq analysis using IgG control
Genome: Mouse (mm10)

Link to the data: [MolBio_project]

Background
Chromatin structure regulates gene expression through histone modifications. Two important histone marks analyzed in this experiment are:

Histone Mark	Functional Role
H3K4me3	Enriched at transcription start sites (TSS); indicates active promoters
H3K27ac	Marks active enhancers and promoters; associated with transcriptional activation

An **IgG control sample** is included to measure background noise and non-specific antibody binding.

Each student has been assigned:

- One **histone modification dataset (ChIP)**
- One **IgG control dataset**

All datasets contain **1 million paired-end reads**.

Student ID	Condition	Histone Marker	Biological Role	Sample ID	FASTQ Base Name	Control (IgG)	IgG Sample
2021113002	High	H3K4me3	Promoter marker	S294	IL7-H_H3k4me3_S294	IgG High	S286
2022113010	High	H3K27ac	Enhancer marker	S295	IL7-H_H3k27ac_S295	IgG High	
2025812001	Low	H3K4me3	Promoter marker	S298	IL7-Low_H3k4me3_S298	IgG Low	S288
2025711001	Low	H3K27ac	Enhancer marker	S299	IL7-_Low_H3k27ac_S299	IgG Low	
2025711002	High-400	H3K4me3	Promoter marker	S296	Il7_H400_H3k4me3_S296	IgG High-400	S287
2022102013	High-400	H3K27ac	Enhancer marker	S297	IL7-H400_H3k27ac_S297	IgG High-400	
2022113003	Low-400	H3K4me3	Promoter marker	S300	IL7-L-400_H3k4me3_S300	IgG Low-400	S289
2022113008	Low-400	H3K27ac	Enhancer marker	S301	IL7-L-400_H3k27ac_S301	IgG Low-400	
2022101048	High-400	H3K4me3	Promoter marker	S296	Il7_H400_H3k4me3_S296	IgG High-400	S287
2021113001	High-400	H3K27ac	Enhancer marker	S297	IL7-H400_H3k27ac_S297	IgG High-400	
2022711001	High	H3K4me3	Promoter marker	S294	IL7-H_H3k4me3_S294	IgG High	S286
Student12	High	H3K27ac	Enhancer marker	S295	IL7-H_H3k27ac_S295	IgG High	

Assignment Questions

1. Raw Read Quality Assessment

Run FastQC on the raw FASTQ files.

Questions

1. What is the average per-base sequence quality across the reads?
2. Are there any positions where sequencing quality decreases significantly?
3. Is there evidence of adapter contamination in the reads?
4. Why is quality control important before downstream analysis?

2. Adapter Trimming

Adapter sequences should be removed using **Trim Galore**.

Questions

1. Why must adapter sequences be removed before alignment?
2. What could happen if adapters are not removed from sequencing reads?
3. What quality threshold was used for trimming in this experiment?

3. Alignment to Reference Genome

Reads were aligned to the **mouse reference genome (mm10)** using Bowtie2.

Questions

1. What percentage of reads aligned successfully to the genome?
2. What does the term **properly paired reads** mean in paired-end sequencing?
3. Why is the **maximum fragment length (-X parameter)** specified during alignment?
4. What biological or technical reasons might cause reads to fail alignment?

4. BAM Processing and Duplicate Removal

PCR duplicates were removed after alignment.

Questions

1. What are PCR duplicates and how do they arise during library preparation?
2. Why can PCR duplicates create false signals in ChIP-seq experiments?
3. What is the purpose of fixing mate information before duplicate removal?

5. Filtering Low-Quality Alignments

Low mapping quality reads were filtered using a MAPQ threshold.

Questions

1. What does **MAPQ score** represent in sequence alignment?
2. Why is a MAPQ ≥ 20 threshold commonly used?
3. What type of genomic regions typically produce low MAPQ reads?

6. Peak Calling with MACS2

Peaks were identified using **MACS2**.

Questions

1. What does a “peak” represent in a ChIP-seq experiment?
2. Why is a control sample required for peak calling?
3. How does MACS2 distinguish real enrichment from background noise?

7. Comparison Between Histone Mark and IgG Control

Compare the peak results from your histone mark with the IgG dataset.

Questions

1. How many peaks were identified in your histone mark dataset?
2. How many peaks were detected in the IgG control sample?
3. Why should IgG datasets contain significantly fewer peaks?

8. Genomic Distribution of Peaks

Annotate peaks using ChIPseeker.

Questions

1. What percentage of peaks fall in **promoter regions**?
2. What percentage of peaks occur in **intergenic regions**?
3. Why might enhancer-associated marks appear farther from gene promoters?

9. Biological Interpretation of Histone Marks

Questions

1. How does the genomic distribution of **H3K4me3** differ from **H3K27ac**?
2. Why is H3K4me3 strongly associated with transcription start sites?
3. What regulatory role do H3K27ac-marked enhancers play in gene expression?

10. Visualization of ChIP-seq Signal

Visualize the bigWig coverage track in **IGV Genome Browser**.

Questions

1. What does the height of a peak represent in the genome browser?
2. Why are peaks sharper in H3K4me3 datasets compared to H3K27ac datasets?

3. How does the IgG signal differ from histone mark signals in IGV?

11. Interpretation of Experimental Results

Questions

1. What biological conclusions can be drawn from promoter-associated histone marks?
2. How do enhancer histone marks contribute to gene regulation?
3. Why are chromatin modifications considered epigenetic regulators?

Submission Requirements

Students should submit:

- Peak count summary table
- Genome browser screenshot showing one representative peak
- A short explanation of their results (1–2 pages)